



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Computer Science and Engineering Academic Year 2025 – 2026 (Odd Semester)

Degree, Semester & Branch: V Semester B.E Computer Science and Engineering
Course Code & Title: CB3491 Cryptography and Cyber Security
Name of the Faculty member: Mrs. P.Sabeena Burvin, AP/CSE

Innovative Practice Description

- **Unit / Topic:** Unit III / Key distribution and Key management
- **Course Outcome:** CO3
- **Topic Learning Outcome:** TLO 10
- **Activity Chosen:** Collaborative Learning
- **Date Implementation:** 09.09.2025
- **Justification:**

Collaborative learning is an instructional approach in which learners work together in pairs or small groups to achieve shared academic goals. Implementing this approach is justified on pedagogical, cognitive, and social grounds. Collaborative learning can be leveraged to enhance traditional cryptographic key distribution and key management. The main idea is that multiple nodes cooperate to generate, update, or validate cryptographic keys without revealing sensitive data.

- **Time Allotted for the Activity:** 20 minutes
- **Details of the Implementation:**
 - The collaborative learning activity provide students the best opportunities to practice the tasks in an authentic way, including interactions and working with other people.
 - The activity was conducted for the duration of 20 minutes.
 - Initially the teams are formed based on self-selection in which the students are form their own groups of size 2.
 - Once the teams are formed, they are discussed the various key distribution and key management methods.
 - The team members were discussed with them. It is shown in Fig.1.
 - The instructor monitored all the teams and provided any clarification if needed.
 - After 20 minutes of discussion the faculty member called every team to present their Knowledge. It shown in Fig 2.

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PO8	PO9	PO12	PSO1
CO1	3	2	2	2	1	1	1	2

(1 – Low 2 – Moderate 3 – High)

PO / PSO mapping:

Innovative Practice	PO1	PO2	PO3	PO4	PO8	PO9	PO12	PSO1
	3	2	2	2	1	1	1	2
Justification for correlation	Basic Computer Science Engineering and mathematics Knowledge for identifying various key distribution techniques.	Students will be able to analyze and define various types of key distributions techniques.	Students will be able to analyze the key distribution techniques for developing an encryption algorithm.	Students will be able to use the virtual environment for identifying various techniques.	Students will be able to use modern tools to perform the key distribution efficiently.	Students will work effectively as an individual and team for solving the identification process.	Students continuously improve their knowledge in key management process.	Updating the concepts to the current trends are moderately correlated

- **Images / Screenshot of the practice:**



Fig. 1: Team Discussion



Fig 2: Presentation by Anu Subiksha B G

- **Reflective Critique:**
 - ❖ **Feedback of practice from students and other stakeholders:**
 - All the teams actively participated in the activity. The students interact with each other and eagerly learn from their team.
 - ❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)
 - This activity helps the students to get more clarity on the given topics.
 - It helps the students are made to work in a team and share their ideas to others and also improve their communication skills
 - The misunderstood concepts are clarified while the team members are presenting.
 - ❖ **Challenges faced in implementation:**
 - It takes more time to make all the teams present the answers to the given questions.
 - Some of the team members hesitate to present in front of the class.

References:

- <https://www.monash.edu/learning-teaching/TeachHQ/Teachingpractices/collaborative-learning/how-to/collaborative-learning-processes>
- <https://tilt.colostate.edu/TipsAndGuides/Tip/18>

